

ASSIGNMENT

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1. NOTATIONS

Let A be a Noetherian ring. We use the notation $D_{\text{Coh}}(A)$ for the derived category of A -complexes with finitely generated cohomology modules. This can be combined with superscripts to make $D_{\text{Coh}}^b(A)$, $D_{\text{Coh}}^+(A)$, $D_{\text{Coh}}^-(A)$, the last of which being exactly the collection of pseudo-coherent objects [Stacks, Tag 064N].

2. COMPLETE RINGS

Let A be a Noetherian ring, I an ideal of A . Assume that A is I -adically complete (i.e. $A = \lim_n A/I^n$). Prove the following theorem.

Theorem 2.1 (this holds much more generally: see [Stacks, Tag 091T]). *Let M be a finitely generated A -module (in particular I -adically complete). Then M is derived I -complete.*

(Hint: we may assume $I = (f)$. Use the explicit resolution of A_f given on Sept 15 to calculate $R\text{Hom}_A(A_f, M)$. If this is too cumbersome due to f^∞ -torsion of M , reduce to the case f is a nonzerodivisor on M , cf. [Stacks, Tag 00L0].)

Alternatively, show that $\text{Hom}_A(A_f, M)$ and $\text{Ext}_A^1(A_f, M)$ are both zero, using the extension definition of Ext^1 ; this is carried out in [Stacks, Tag 091R].)

3. TESTING BOUNDED PSEUDO-COHERENT OBJECTS

Let A be a Noetherian ring, I an ideal of A . Assume that A is I -adically complete. Prove the following theorem.

Theorem 3.1 ([Lyu25, Lemma 5.4]). *Let $M \in D^b(A)$ be derived I -complete. If $R\text{Hom}_A(A/I, M) \in D_{\text{Coh}}(A/I)$, then $M \in D_{\text{Coh}}(A)$.*

(Hint: as seen on Sept 15 if $R\text{Hom}_A(A/I, M) = 0$ then $M = 0$. Using some similar ideas and try to resolve M from the right.)

REFERENCES

- [Lyu25] Shiji Lyu. “Formal lifting of dualizing complexes and consequences”. In: *Nagoya Math. J.* 258 (2025), pp. 219–236.
- [Stacks] The Stacks Project authors. *The Stacks Project*. URL: <https://stacks.math.columbia.edu/>.

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